M.U. Faculty of Engineering
Fall 2020 (2020-2021) **December 10, 2020**

IE381/3081 Modeling and Discrete Simulation
Midterm Exam

Important Notice

- Points for each question are shown inline with the question next to the question number.
- Notes, slides, textbooks and other published materials are open. You may use calculators.
- The use of the Internet (either via computer/notebook or mobile phone) is not allowed.

Please follow the following steps in answering the questions and submitting your solution

1. Print this exam document and write your answers with your handwriting clearly and neatly. Show your work in answering the questions.
2. Fill the name, student ID and then sign at the top of **each page**.
3. If you cannot print this exam file,
   - Solve all questions in your handwriting on blank A4 sheets
   - Write the sentences below related to the honor code to the top of **each page** among with your name, student ID, date and then sign.
4. Scan all solution pages into a single file in PDF, and name it as "FirstName_LastName.pdf"
5. Upload your PDF file to the https://ues.marmara.edu.tr/ before deadline.

I hereby pledge and swear that I have not violated the honor code and the work done is totally my own. On my honor, I have neither given nor received nor observed any unauthorised aid or assistance on this exam.

Name Surname: _____

ID : _____

Signature : _____

Questions	1	2	Total Points
Points	40	60	
Points you get			

Name: _____ **ID:** _____ **Signature**_____

General Formulas for Queue Systems

$$\rho = \frac{\lambda}{\mu}$$

$$\hat{L} = \hat{\lambda}\hat{w}$$

$$L = \lambda w \quad \text{as} \quad T \to \infty \text{ and } N \to \infty$$

$$\hat{L} = \frac{1}{T}\sum_{i=0}^{\infty} iT_i = \sum_{i=0}^{\infty} i\left(\frac{T_i}{T}\right) \to L \quad \text{as} \quad T \to \infty$$

$$\hat{L}_Q = \frac{1}{T}\sum_{i=0}^{\infty} iT_i^Q \to L_Q \quad \text{as} \quad T \to \infty$$

$$\hat{w} = \frac{1}{N}\sum_{i=1}^{N} W_i \to w \quad \text{as} \quad N \to \infty$$

$$\hat{w}_Q = \frac{1}{N}\sum_{i=1}^{N} W_i^Q \to w_Q \quad \text{as} \quad N \to \infty$$

Poisson Distribution

Probability mass function

$$p(x) = \begin{cases} \dfrac{e^{-\alpha}\alpha^x}{x!}, & x = 0, 1, \ldots \\ 0, & \text{otherwise} \end{cases}$$

Cumulative Distribution Function

$$F(x) = \sum_{i=0}^{x} \frac{e^{-\alpha}\alpha^x}{i!}$$

Mean : $E(x) = \alpha = n.p$

Variance : $\sigma^2 = \alpha$

Expected value of X (*mean of X*)

If *X* is discrete:

$$E(x) = \sum_{\text{all } i} x_i p(x_i)$$

If *X* is continuous:

$$E(x) = \int_{-\infty}^{\infty} xf(x)dx$$

Steady-State Parameters of M/G/1 Queue

$$\rho = \lambda/\mu, \quad P_0 = 1 - \rho$$

$$L = \rho + \frac{\rho^2(1+\sigma^2\mu^2)}{2(1-\rho)}, \quad L_Q = \frac{\rho^2(1+\sigma^2\mu^2)}{2(1-\rho)}$$

$$w = \frac{1}{\mu} + \frac{\lambda(1/\mu^2 + \sigma^2)}{2(1-\rho)}, \quad w_Q = \frac{\lambda(1/\mu^2 + \sigma^2)}{2(1-\rho)}$$

Steady-State Parameters of M/M/1 Queue

$$\rho = \lambda/\mu, \quad P_n = (1-\rho)\rho^n$$

$$L = \frac{\lambda}{\mu-\lambda} = \frac{\rho}{1-\rho}, \quad L_Q = \frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{\rho^2}{1-\rho}$$

$$w = \frac{1}{\mu-\lambda} = \frac{1}{\mu(1-\rho)}, \quad w_Q = \frac{\lambda}{\mu(\mu-\lambda)} = \frac{\rho}{\mu(1-\rho)}$$

Steady-State Parameters of M/M/c/∞/∞ Queue

$$\rho = \lambda/c\mu$$

$$P_0 = \left\{\left[\sum_{n=0}^{c-1}\frac{(\lambda/\mu)^n}{n!}\right] + \left[\left(\frac{\lambda}{\mu}\right)^c\left(\frac{1}{c!}\right)\left(\frac{c\mu}{c\mu-\lambda}\right)\right]\right\}^{-1}$$

$$L = c\rho + \frac{(c\rho)^{c+1}P_0}{c(c!)(1-\rho)^2} = c\rho + \frac{\rho\, P(L(\infty) \geq c)}{1-\rho}$$

$$w = \frac{L}{\lambda}$$

Steady-State Parameters of M/M/c/K/K Queue

$$P_0 = \left\{\sum_{n=0}^{c-1}\binom{K}{n}\left(\frac{\lambda}{\mu}\right)^n + \sum_{n=c}^{K}\frac{K!}{(K-n)!c!c^{n-c}}\left(\frac{\lambda}{\mu}\right)^n\right\}^{-1}$$

$$P_n = \begin{cases} \binom{K}{n}\left(\frac{\lambda}{\mu}\right)^n P_0, & n = 0,1,\ldots,c-1 \\ \dfrac{K!}{(K-n)!c!c^{n-c}}\left(\frac{\lambda}{\mu}\right)^n, & n = c, c+1, \ldots K \end{cases}$$

$$L = \sum_{n=0}^{K} nP_n, \quad w = L/\lambda_e, \quad \rho = \lambda_e/c\mu$$

$$\lambda_e = \sum_{n=0}^{K}(K-n)\lambda P_n$$

Variance of *X* (*also denoted as* σ^2)

$V(X) = E[(X - E[X])^2]$ or

$V(X) = E(X^2) - [E(X)]^2$

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IE381/3081 Modeling and Discrete Simulation, Midterm Exam, 10 December 2020:

Problem 1 [40 points]:

The following probability density function (***pdf***) is given for random variable ***X***, that is the service time (in minutes) of a teller in a bank.

$$f(x) = \begin{cases} \frac{1}{3}, & 0 < x < 1 \\ \frac{2x^2}{7}, & 1 \le x \le 2 \\ 0, & otherwise \end{cases}$$

(a) Draw the graph for the given ***pdf*** for ***X***, *f(x)*.

(b) Find the cumulative distribution function (***cdf***) for ***X***, ***F(x)***.

(c) Draw the graph for the cumulative distribution function (***cdf***) for ***X***, ***F(x)***.

(d) Calculate the ***P(X<1.5).***

Problem 2 [60 points]:

The library has a print service to provide students printing their documents/homeworks. The print orders submitted by the students are sent to the printing service. At the print service, all print requests are queued in the printing queue of a single printer, which the printing service works based on a first-come, first-served priority rule. It is known that 18 print requests per hour (on average) arrive to the printing service. It is also known that arrivals are Poisson distribution. The printing system can help an average of 24 print requests per hour, with the service rate being described by an exponential distribution. Calculate the following operating characteristics of the printing service.

(a) The average utilization of the printing service

(b) The average number of print requests in the system

(c) The average number of print requests waiting in line

(d) The average time a print request spends in the system

(e) The average time a print request spends waiting in line

(f) The probability of having more than 4 print requests in the system